

An Introduction to Firewall Technology



凌群電腦

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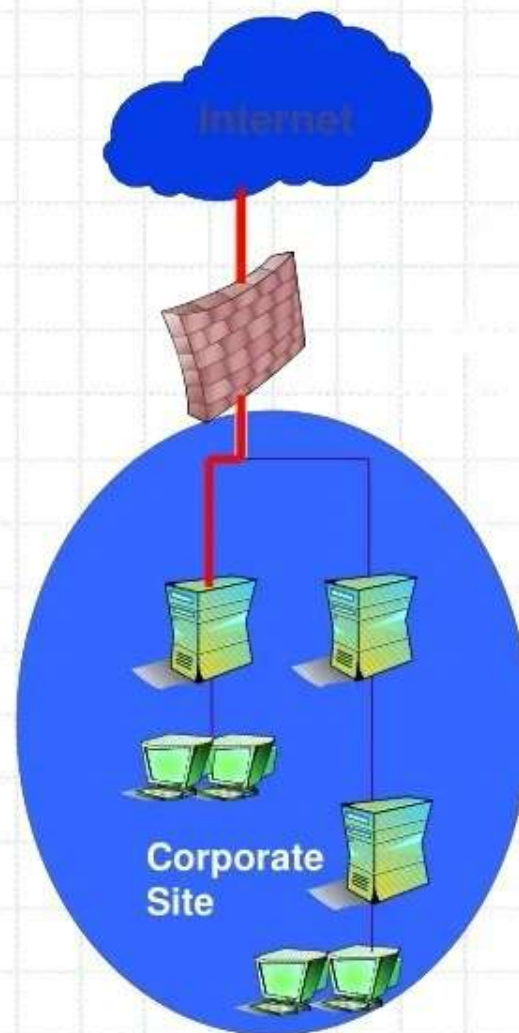
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Agenda

- ◆ What is a firewall
- ◆ Why an organization needs a firewall
- ◆ Types of firewalls and technologies
- ◆ Deploying a firewall
- ◆ What is a VPN

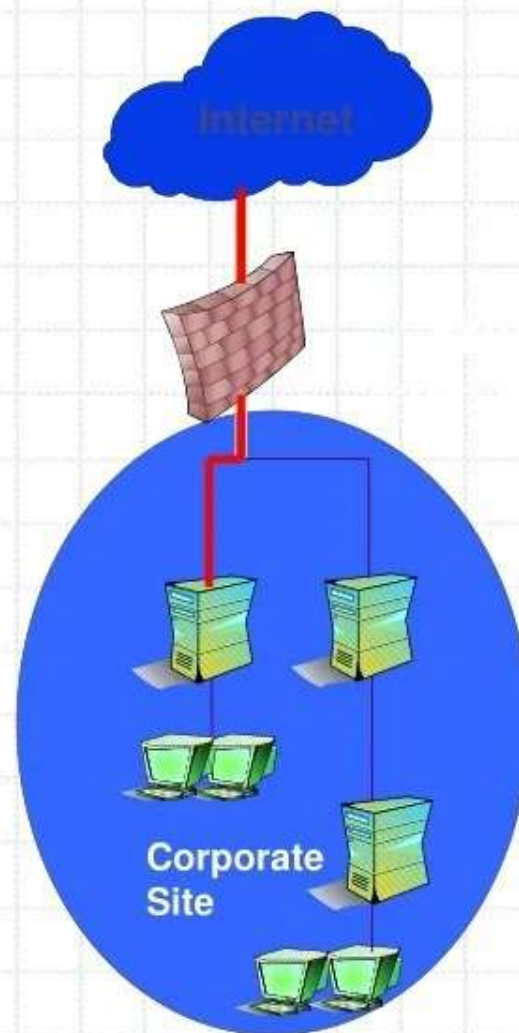
What is a Firewall ?

- ◆ A firewall :
 - Acts as a security gateway between two networks
 - Tracks and controls network communications
 - Decides whether to pass, reject, encrypt, or log communications (Access Control)



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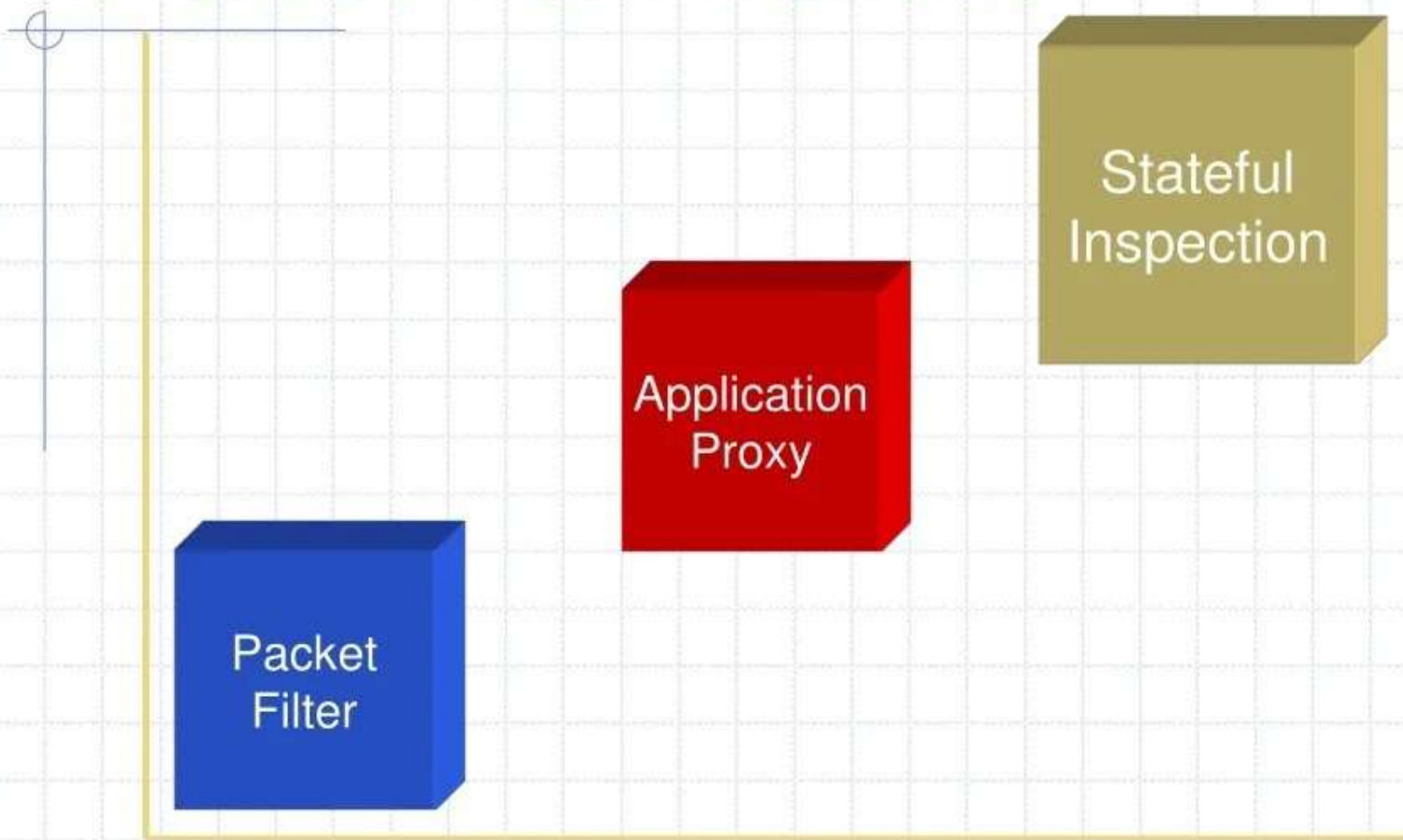
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Why Firewalls are Needed

- ◆ Prevent attacks from untrusted networks
- ◆ Protect data integrity of critical information
- ◆ Preserve customer and partner confidence

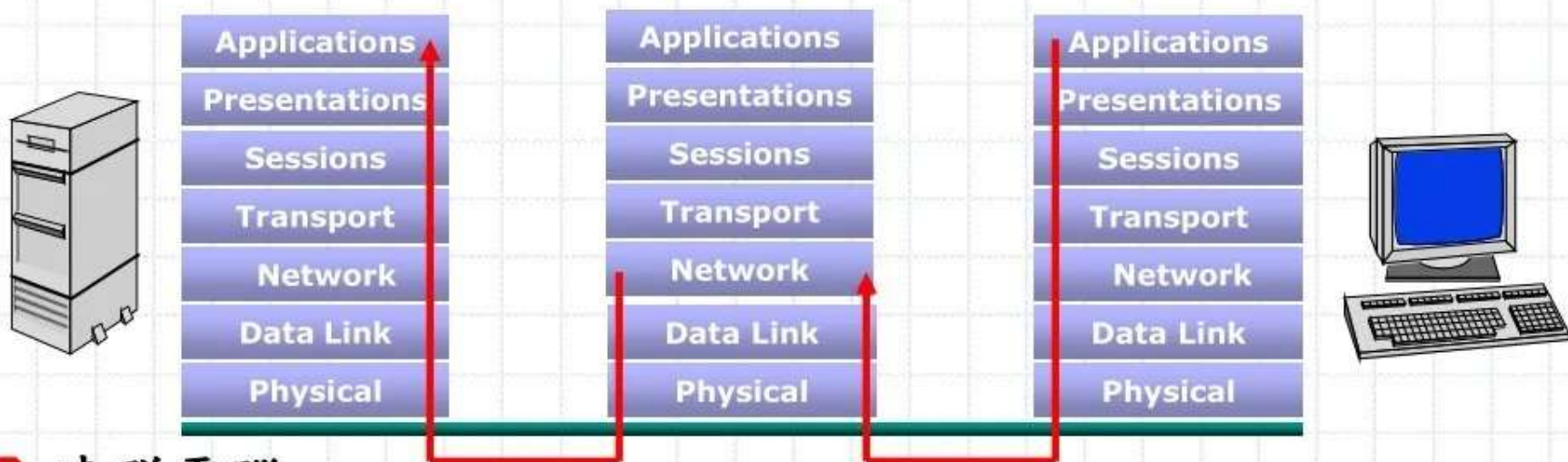
Evolution of Firewalls



Stage of Evolution

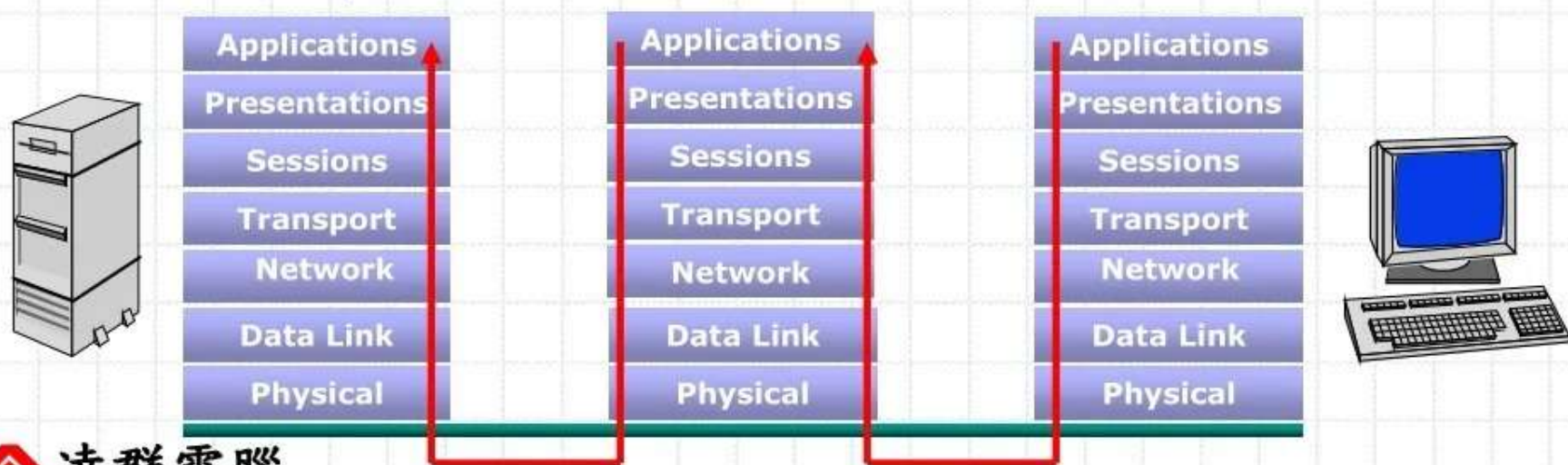
Packet Filter

- ◆ Packets examined at the network layer
- ◆ Useful “first line” of defense - commonly deployed on routers
- ◆ Simple accept or reject decision model
- ◆ No awareness of higher protocol layers



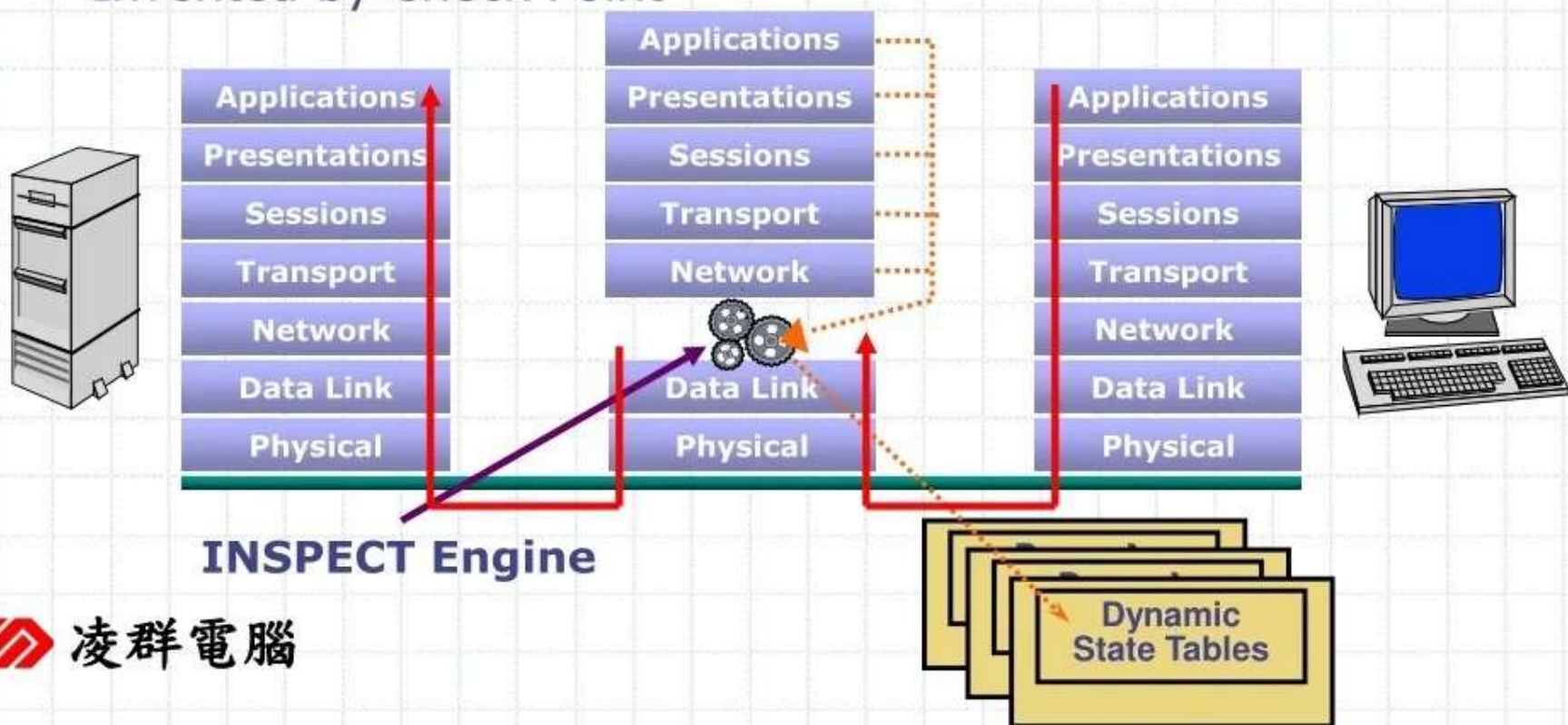
Application Gateway or Proxy

- ◆ Packets examined at the application layer
- ◆ Application/Content filtering possible - prevent FTP “put” commands, for example
- ◆ Modest performance
- ◆ Scalability limited



Stateful Inspection

- ◆ Packets Inspected between data link layer and network layer in the OS kernel
- ◆ State tables are created to maintain connection context
- ◆ Invented by Check Point

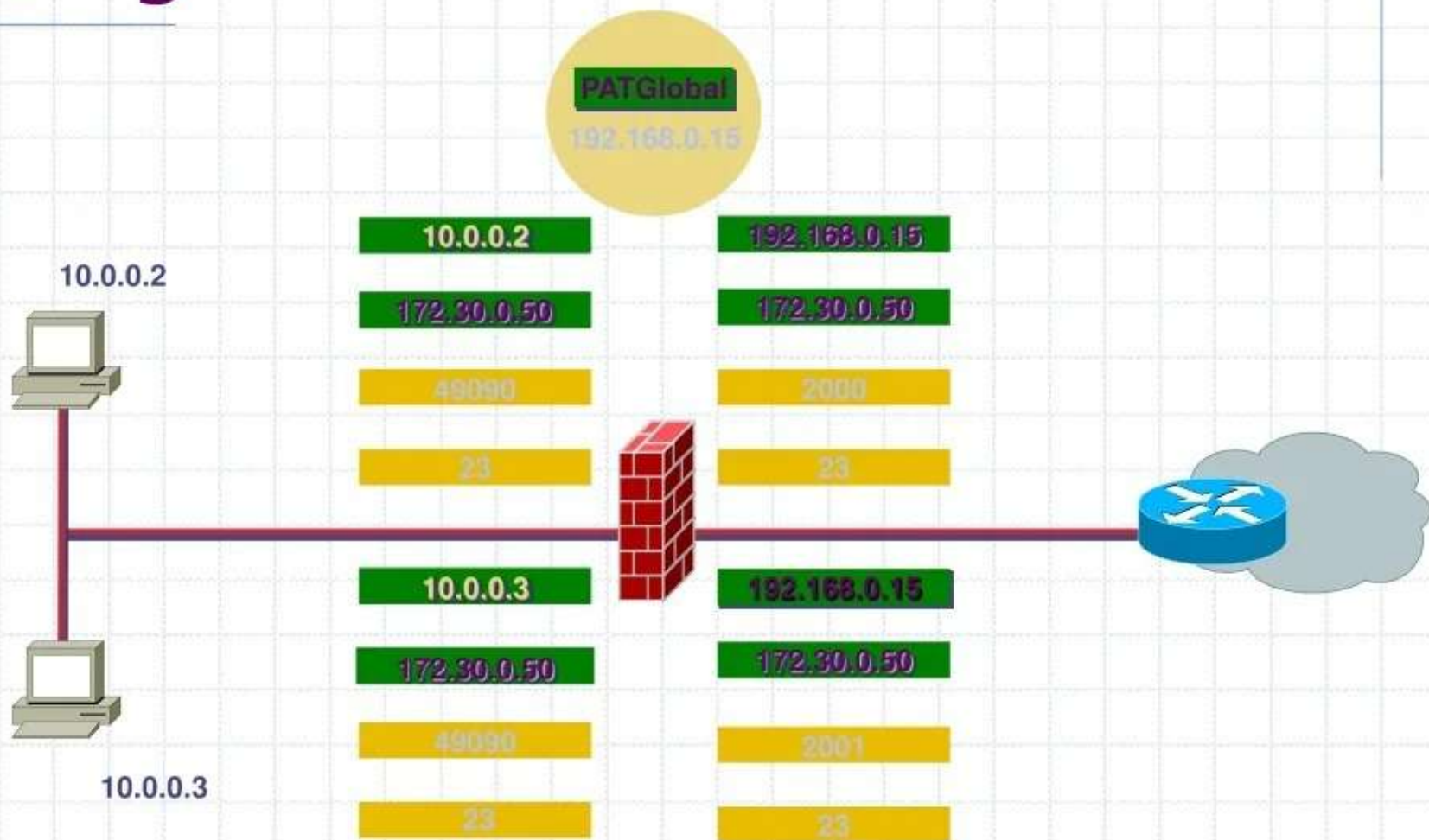


Network Address Translation (NAT)



- ◆ Converts a network's illegal IP addresses to legal or public IP addresses
 - Hides the true addresses of individual hosts, protecting them from attack
 - Allows more devices to be connected to the network

Port Address Translation—Hiding



Personal Firewalls

- ◆ Need arises from always on connections
- ◆ Your PC is not protected enough by your OS
- ◆ Intrusion detection facilities
- ◆ Different levels of security
 - Templates

Firewall Deployment

- ◆ Hardware appliance based firewall
 - Single platform, software pre-installed
 - Can be used to support small organizations or branch offices with little IT support
- ◆ Software based firewall
 - Flexible platform deployment options
 - Can scale as organization grows

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Summary

- ◆ Firewalls foundation of an enterprise security policy
- ◆ Stateful Inspection is the leading firewall technology

選擇防火牆參考依據

- ◆ 防火牆必須允許/拒絕的網路協定或應用層網路傳輸？
- ◆ 防火牆在控制網路傳輸時是否需要作使用者身份認證？
- ◆ 如何建立規則？
- ◆ 是否可隱藏網址？
- ◆ 是否有一個以上的網址，能夠保護網路上數個 **web** 和 **email** 伺服器不受攻擊？

選擇防火牆參考依據 續

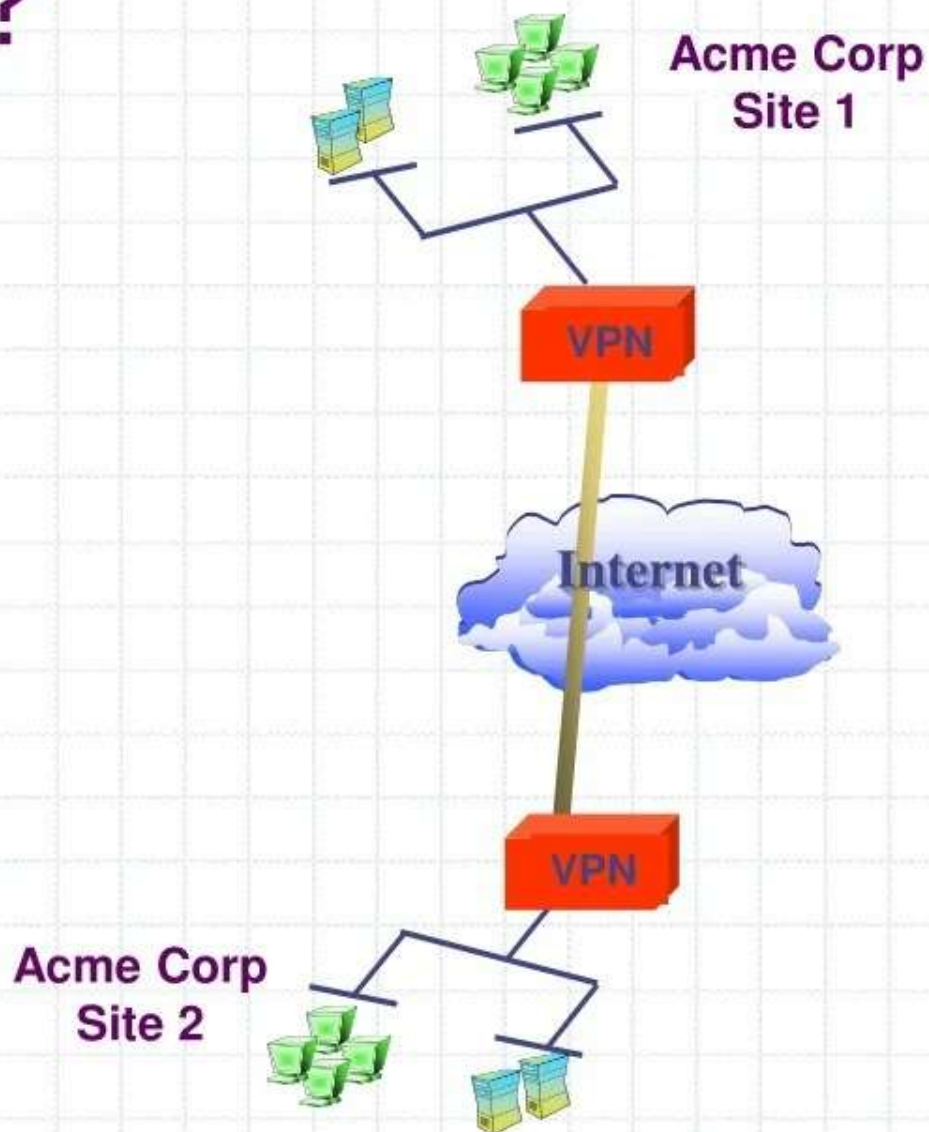
- ◆ 是否可過濾 Java 和 ActiveX？
- ◆ 它如何強化作業系統安全？
- ◆ 是否在不影響安全性的情況下處理所有的網路傳輸活動？
- ◆ 是否提供事件紀錄和警告？
- ◆ 是否簡單易用？
- ◆ 是否支援附加其他的事件報告軟體？
- ◆ 是否提供內容阻擋功能？

選擇防火牆參考依據 續

- ◆ 是否具擴充性，以符合未來的需求？
- ◆ 是否易於加入遠端的防火牆和行動使用者？
- ◆ 是否和市面上其他產品互通？

What is a VPN?

- ◆ A VPN is a private connection over an open network
- ◆ A VPN includes authentication and encryption to protect data integrity and confidentiality



Why Use Virtual Private Networks?

- ◆ More flexibility
- ◆ More scalability
- ◆ Lower costs
 - Reduced frame relay/leased line costs
 - Reduced long distance
 - Reduced equipment costs (modem banks, CSU/DSUs)
 - Reduced technical support

Why Use Virtual Private Networks?

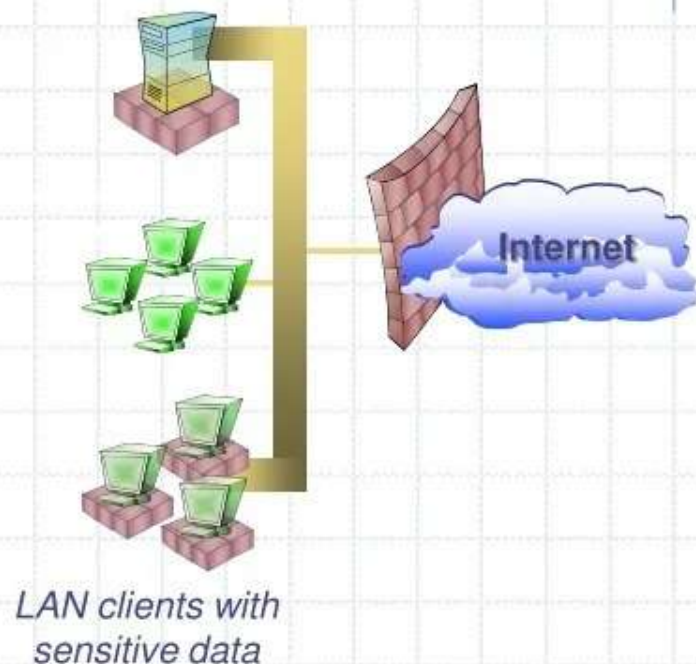
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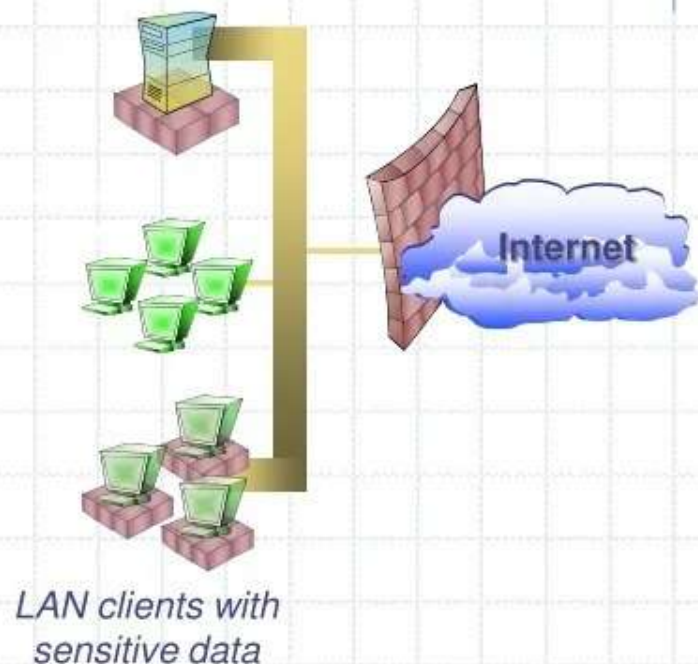
Types of VPNs

- ◆ Remote Access VPN
- ◆ Site-to-Site VPN
- ◆ Extranet VPN
- ◆ Client/Server VPN
 - Protects sensitive internal communications



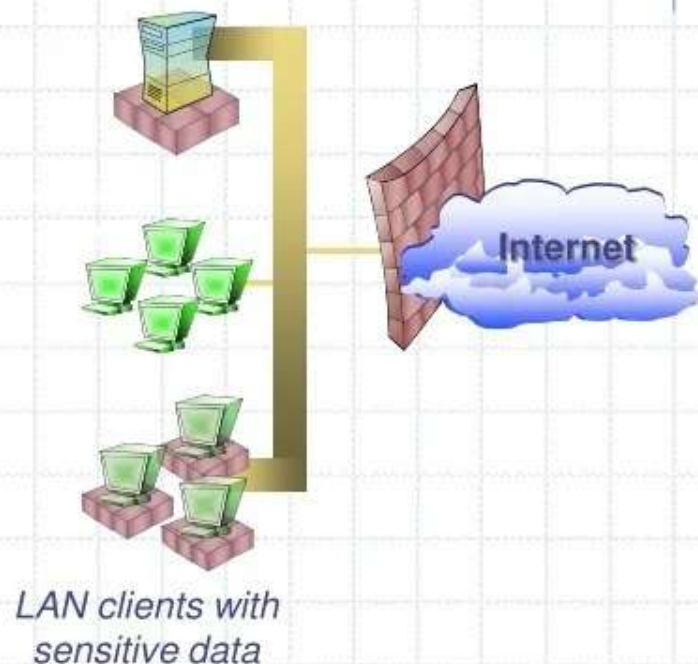
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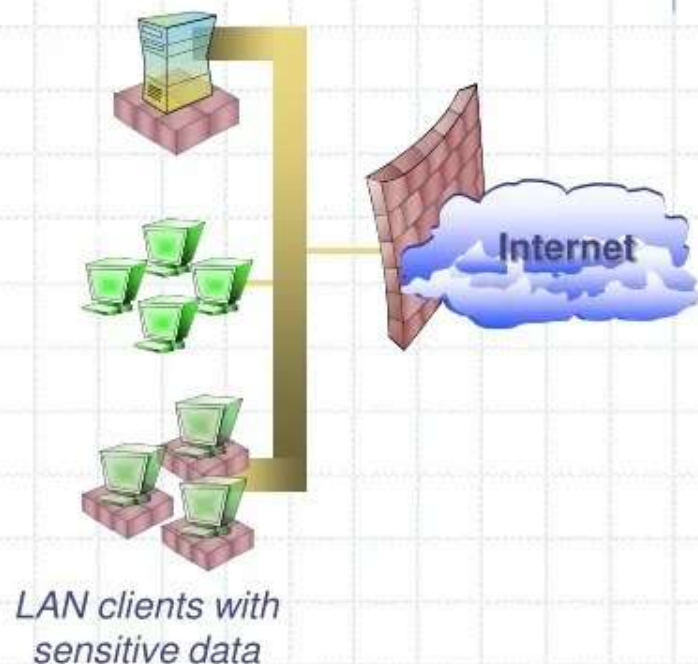
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Components of a VPN

- ◆ Encryption
- ◆ Key management
- ◆ Message authentication
- ◆ Entity authentication

Encryption



- ◆ Current standards: DES and Triple-DES
 - Over 20 years in the field
- ◆ AES beginning deployment
 - New standard
 - More computationally efficient
 - Longer keys = more secure

Key Management

- ◆ Public key cryptosystems enable secure exchange of private crypto keys across open networks
- ◆ Re-keying at appropriate intervals
- ◆ IKE = Internet Key Exchange protocols
 - Incorporates ISAKMP/Oakley

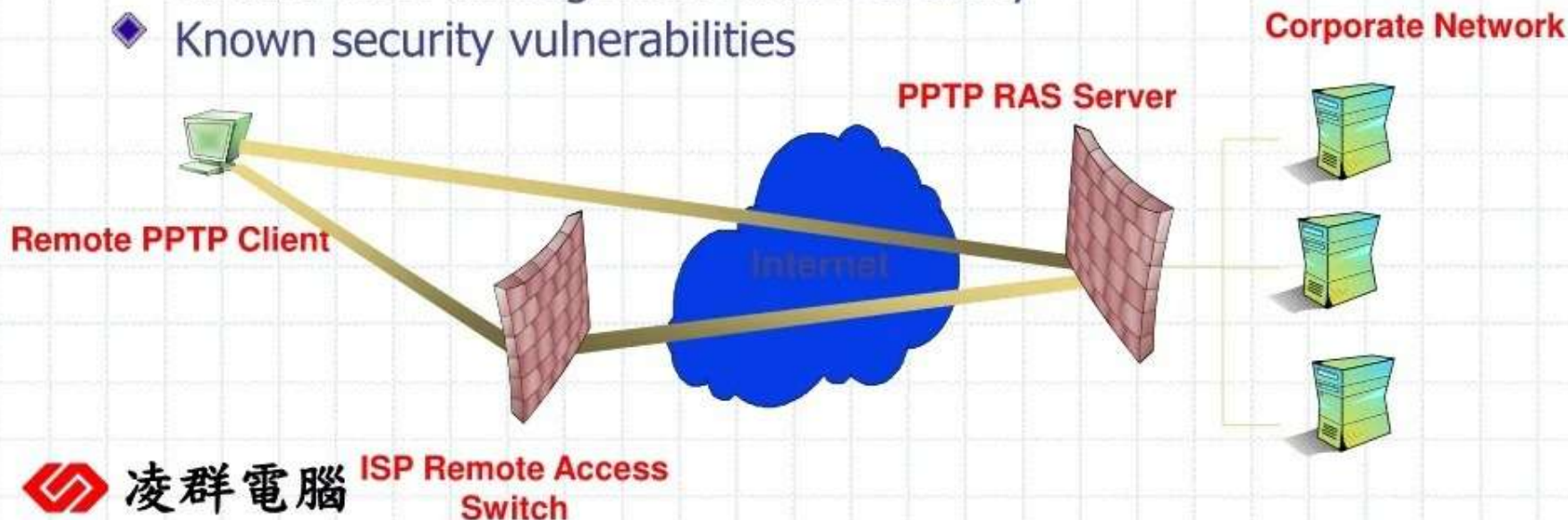


Authentication

- ◆ IPsec standards focus on authentication of two network devices to each other
 - IP address/preshared key
 - Digital certificates
- ◆ User authentication is added on top if required
 - RADIUS and TACACS+ are the standard protocols for authentication servers
- ◆ XAUTH is being added to the standards to address user authentication

Point-to-Point Tunneling Protocol

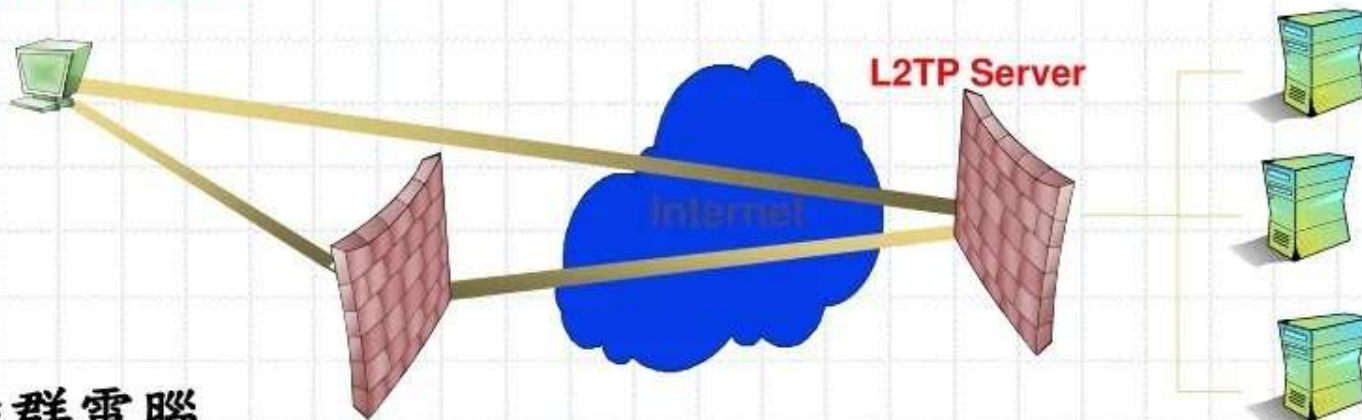
- ◆ Layer 2 remote access VPN distributed with Windows product family
 - Addition to Point-to-Point Protocol (PPP)
 - Allows multiple Layer 3 Protocols
- ◆ Uses proprietary authentication and encryption
- ◆ Limited user management and scalability
- ◆ Known security vulnerabilities



Layer 2 Tunneling Protocol (L2TP)

- ◆ Layer 2 remote access VPN protocol
 - Combines and extends PPTP and L2F (Cisco supported protocol)
 - Weak authentication and encryption
 - Does not include packet authentication, data integrity, or key management
 - Must be combined with IPSec for enterprise-level security

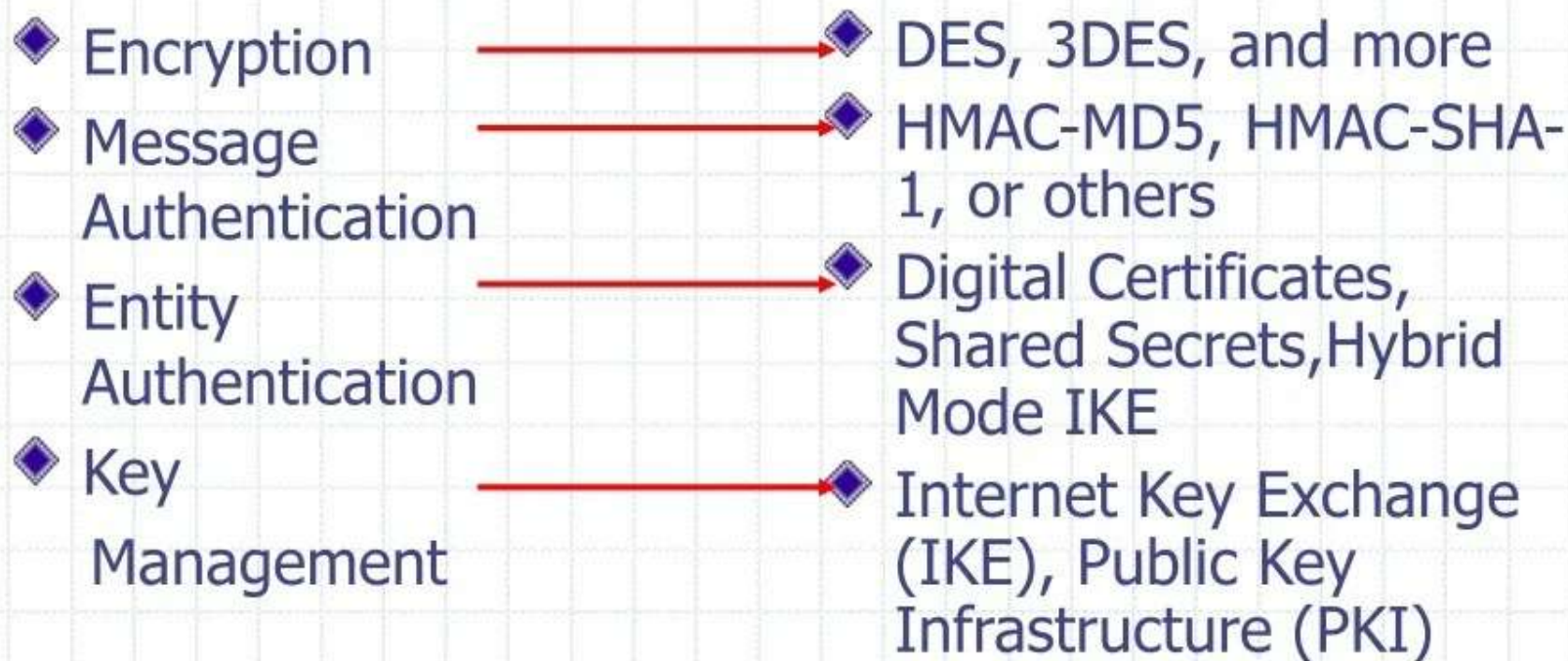
Remote L2TP Client



Internet Protocol Security (IPSec)

- ◆ Layer 3 protocol for remote access, intranet, and extranet VPNs
 - Internet standard for VPNs
 - Provides flexible encryption and message authentication/integrity
 - Includes key management

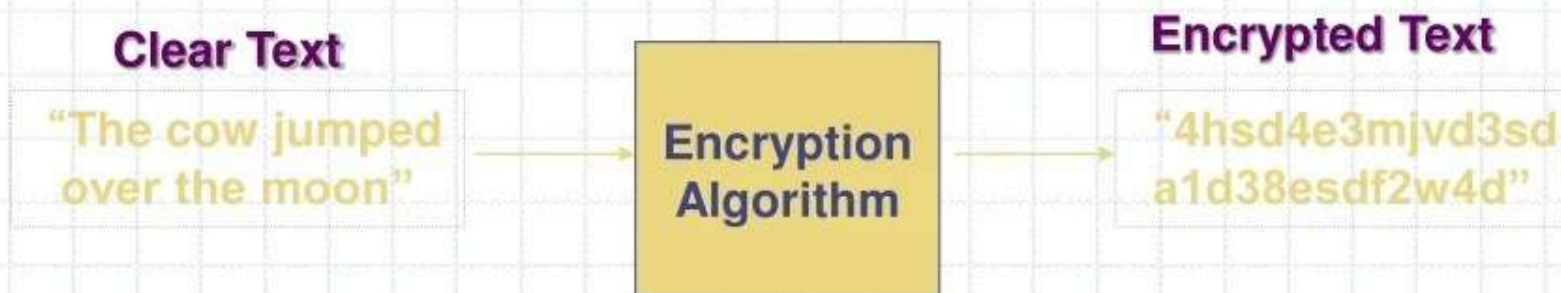
Components of an IPSec VPN



All managed by security associations (SAs)

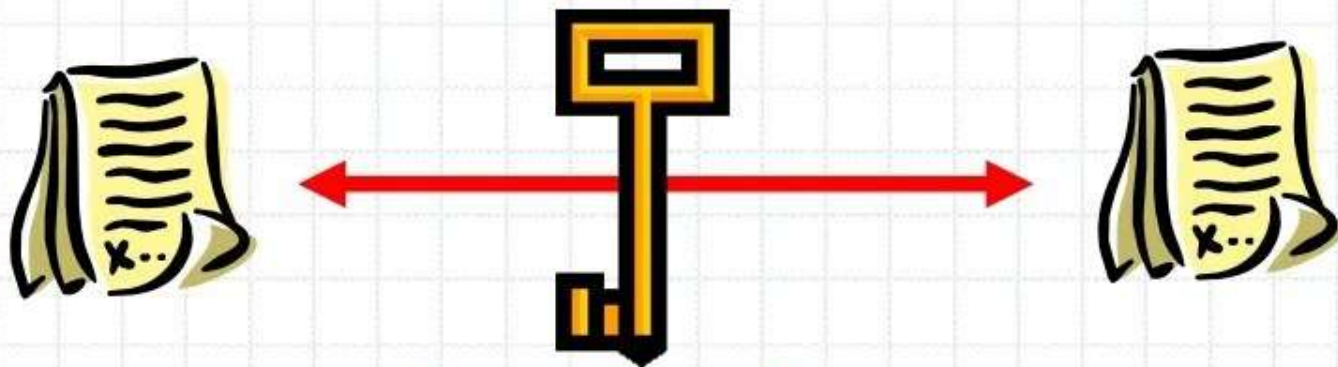
Encryption Explained

- ◆ Used to convert data to a secret code for transmission over an untrusted network



Symmetric Encryption

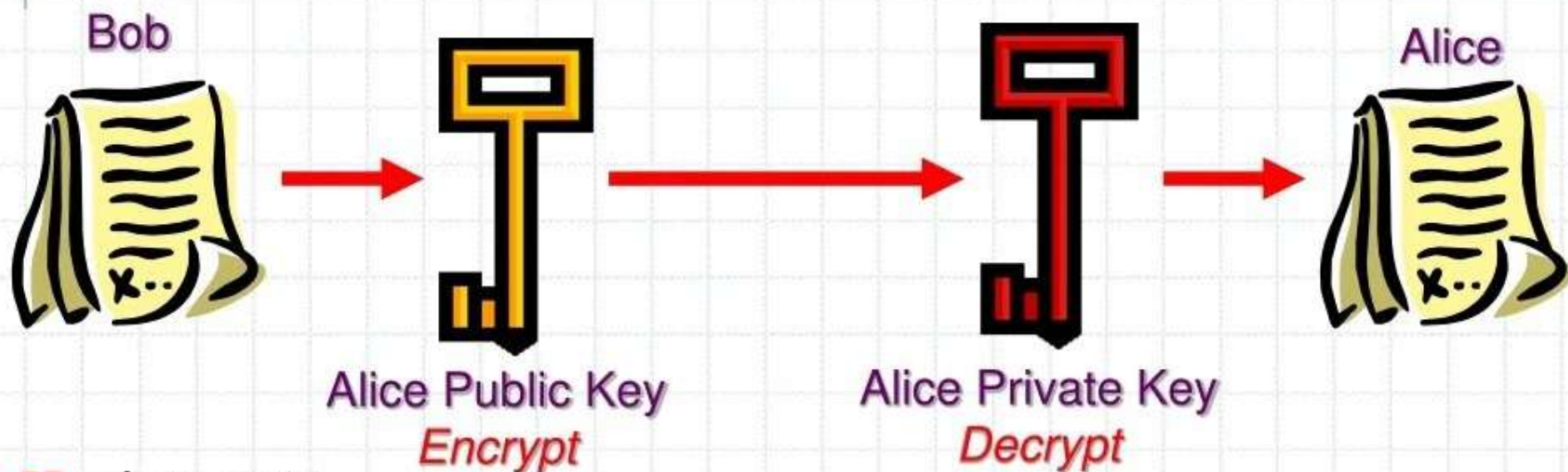
- ◆ Same key used to encrypt and decrypt message
- ◆ Faster than asymmetric encryption
- ◆ Examples: DES, 3DES, RC5, Rijndael



Shared Secret Key

Asymmetric Encryption

- ◆ Different keys used to encrypt and decrypt message (One public, one private)
- ◆ Examples include RSA, DSA, SHA-1, MD-5



目前Internet上使用的加密系統比較表

DNSSEC	安全的網域名稱系統	RSA、MD5	認證性、整體性
IPSec	對IP封包進行加密的低階協定	Diffie-Hellman、DES、3DES、RC4、IDEA	保密性、認證性、整體性
Kerberos	高階程式的網路安全服務	DES	保密性、認證性
SSH	具加密功能的遠端終端機	RSA、Diffie-Hellman、DES、3-DES、Blowfish、其它	保密性、認證性

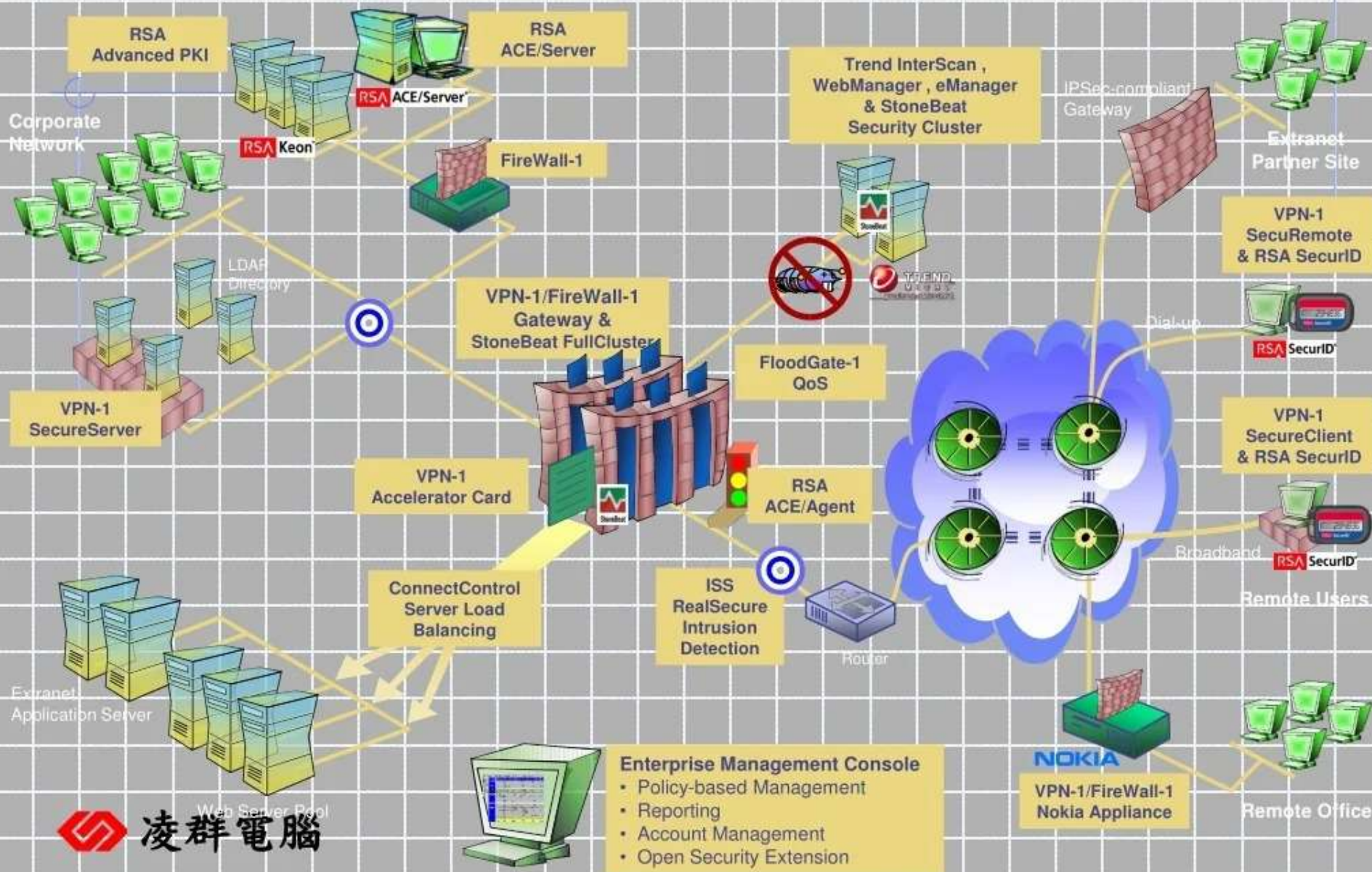
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破解 DES Keys 的成本時間參考

攻擊型態	花費預算	40-Bit	56-Bit	168-Bit 3-DES
個人駭客	美金 400	5 小時	38 年	無法預計
專業駭客	美金 1萬	12 分鐘	556 日	10^{19} 年
智慧型組織	美金 1千萬	0.02 秒	21 分鐘	10^{17} 年

Secure Virtual Network Architecture



Thank You!

